

$$\begin{array}{c}^{12}\\ \Omega\\ \Omega\\ \dot{t}\\ \dot{\theta}\\ \dot{\theta}\\ \dot{r}_0\\ g\\ R\\ \Theta\end{array}$$

$$(1)\quad \{ \, drdt = \dot{r}, d\theta dt = \dot{\theta}, d\dot{r}dt = R + r(\dot{\theta} \pm \Omega)^2 - g\,(r_0r)^2, rd\dot{\theta}dt = \Theta - 2\dot{r}(\dot{\theta} \pm \Omega) .$$

$$\begin{array}{c}radial_a cc\\ R\end{array}$$

$$\begin{array}{c}\Theta\\ S\\ L\\ D\\ W\\ g\\ V\\ \Sigma, A\\ \Delta\end{array}$$

$$(2)\quad \Sigma = \frac{Sg}{W}, \Lambda = \frac{Lg}{WV}, \Delta = \frac{Dg}{WV}.$$

$$\begin{array}{c}w\\ w\\ w\\ v_r\\ v_\theta\end{array}$$

$$(3)\quad \{ \, v_{\,r} = \dot{r}, v_\theta = r\dot{\theta},$$

$$(4)\quad V^2 = \dot{r}^2 + (r\dot{\theta} + w)^2.$$

$$\begin{array}{c}\beta\\ R\\ \Theta\end{array}$$

$$(5)\quad \{ R = \Sigma \sin \beta + (v_\theta + w) \Lambda - v_r \Delta, \Theta = \Sigma \cos \beta - v_r \Lambda - (v_\theta + w) \Delta.$$

$$\begin{array}{c}N\\ \frac{d\beta}{dt} = \frac{d\dot{\theta}}{dt} + N.\end{array}$$

$$(6)$$

$$\begin{array}{c}L\\ D\\ m\\ C_L^L\\ C_D^L\\ L\\ D\end{array}$$

$$(7)\quad \{ \, L = \frac{1}{2} \rho V^2 A C_L, D = \frac{1}{2} \rho V^2 A C_D,$$

$$\begin{array}{c}\rho\\ r\\ A\\ A\\ \dot{\theta}\\ \gamma\\ \gamma\\ C_L^L\\ C_D^L\\ \alpha\\ \alpha\\ R^e\\ M\\ a\end{array}$$

$$(8)\quad M = \frac{V}{a},$$

$$\begin{array}{c}g\\ \mu\end{array}$$

$$(9)\quad Re = \frac{\rho V l}{\mu}.$$

$$\begin{array}{c}\mu\\ r\end{array}$$

$$(10)\quad \{ \, C_{\,L} = C_L(\alpha, \gamma, M, Re), C_D = C_D(\alpha, \gamma, M, Re).$$

$$\begin{array}{c}m\\ m\\ C_M\end{array}$$

$$(11)\quad m = \frac{1}{2} \rho V^2 A l C_M.$$

$$\begin{array}{c}C_M\\ \alpha, \gamma, M\\ R^e\\ C_M = C_M(\alpha, \gamma, M, Re).\end{array}$$